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Lesson 3: Regression

Technologies for Artificial Intelligence

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Before we start

- Exercise 2, which we discuss now
- The pipeline you built is the one we fit models into from now on

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Today: the first real model

- The model and what it claims
- The exact solution, and why it sometimes fails
- Gradient descent
- Curves, and the price of flexibility
- Ridge and Lasso

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One dataset, all lesson

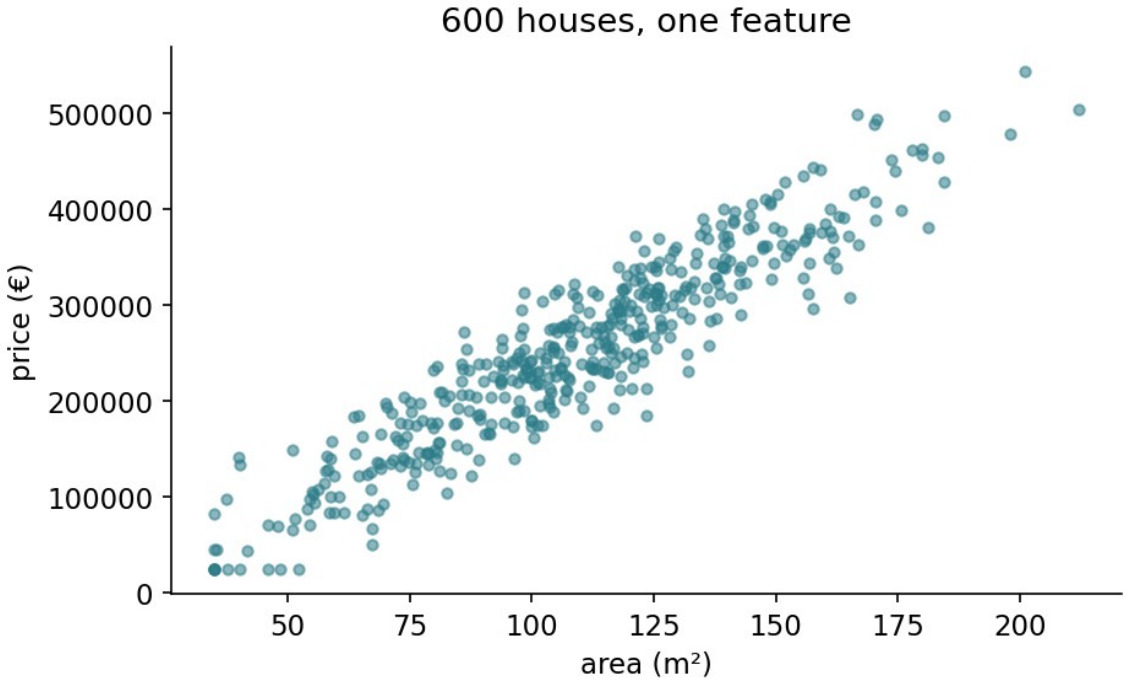
600 houses. Six measurements. A price.

And we know the coefficients that generated it.

In code: `train_test_split(X, y, test_size=0.25)` keeps 150 houses for testing.

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What we are trying to predict



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What a linear model claims

Each feature contributes a fixed amount per unit, and the contributions add up: $\hat{y} = w_1x_1 + w_2x_2 + \ldots + w_nx_n + b$, one coefficient w per feature plus an intercept b.

- Every square metre: the same 2,400 €
- Every kilometre out: the same −6,500 €

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Fitting means minimising something

We need one number saying how badly a candidate model is doing.

Adding up the errors, $\Sigma_i (\hat{y}^{(i)} - y^{(i)})$, fails: +50,000 and −50,000 cancel to zero.

Every error must count as a positive amount:

- absolute error $\Sigma_i |\hat{y}^{(i)} - y^{(i)}|$
- squared error $\Sigma_i (\hat{y}^{(i)} - y^{(i)})^2$

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Why squared, not absolute

- Squared error $\Sigma_i (\hat{y}^{(i)} - y^{(i)})^2$: each error $e = \hat{y}^{(i)} - y^{(i)}$ counts as e^2, not as $|e|$
- **Differentiable everywhere**: e^2 is smooth, $|e|$ has a corner at $e = 0$
- **Punishes one large error more**: double an error and $|e|$ doubles, e^2 quadruples
- Under Gaussian noise it is **maximum likelihood**: it picks the coefficients that make the observed prices most probable

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The cost function: mean squared error

$$J(w,b) = \frac{1}{2m} \sum_{i=1}^{m} \left(\hat{y}^{(i)} - y^{(i)}\right)^2$$

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Trying $\hat{y} = 2{,}400 \cdot \text{area} + 45{,}000$

Area (m^2)	True (€)	Predicted (€)	Error $\hat{y} - y$	Error^2
80	240,000	237,000	−3,000	9.0×10^6
120	320,000	333,000	+13,000	1.69×10^8
200	540,000	525,000	−15,000	2.25×10^8

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Squared error spends its attention on the worst house

- The 3,000 € error contributes about **2%** of the total
- The two large errors contribute the other **98%**
- Halving the 15,000 € error cuts the cost by **42%**
- In code: `mean_squared_error(y, y_pred)`, the mean of the squared errors without the ½

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A cost needs something to be compared against

- Trivial answer: **always predict the mean**, `DummyRegressor()`
- Root mean squared error, RMSE = √MSE, on the 150 test houses: **96,440 €** for the mean, **20,341 €** for the model
- $R^2 = 1 - \Sigma(y - \hat{y})^2 \div \Sigma(y - \bar{y})^2 =$ **0.956** (`r2_score`): 0 is the mean, 1 is perfect
- Quote the euros too: R^2 depends on *this* test set's variance

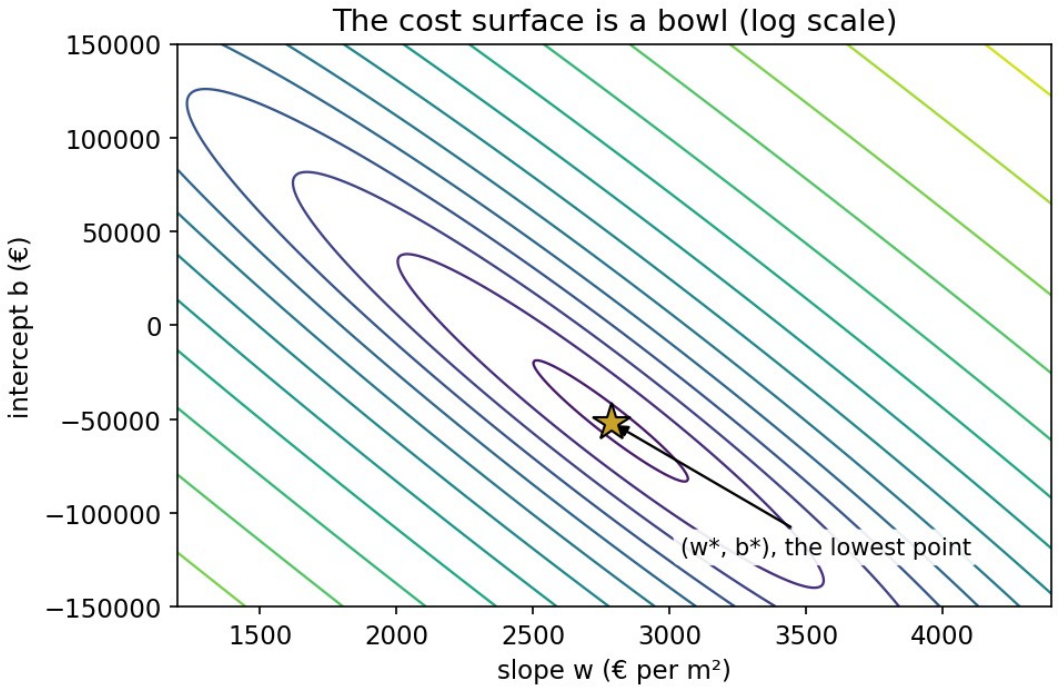
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Fitting is finding where J is smallest

$$(w^*, b^*) = \arg\min_{w,\ b} J(w, b)$$

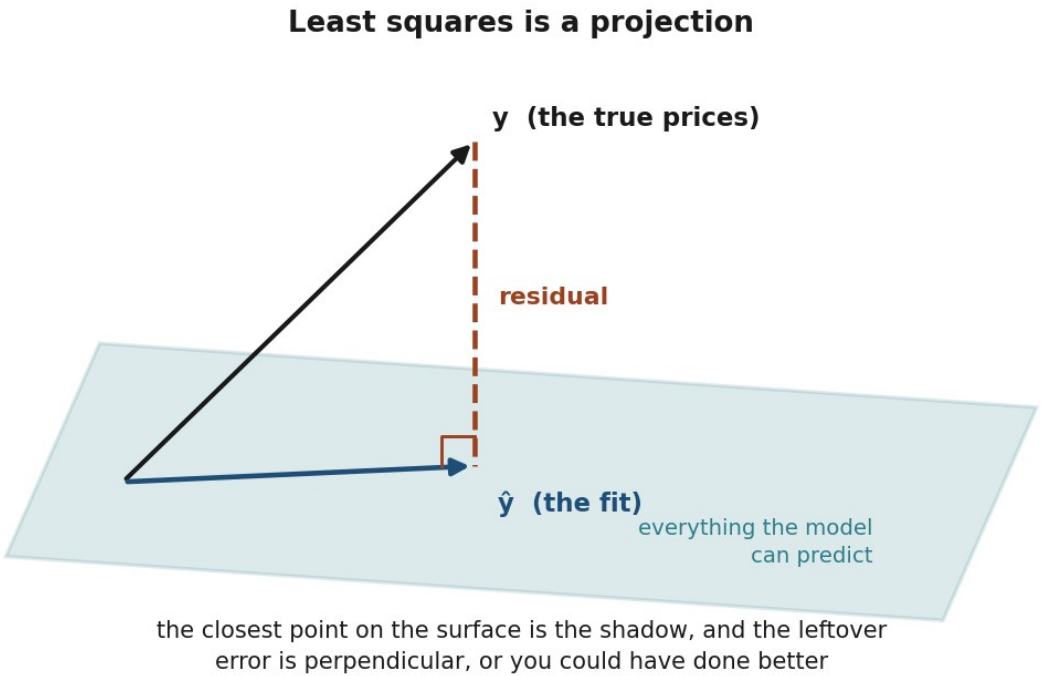
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The cost is a bowl



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The picture behind the exact solution



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θ and X: all parameters, all houses

- $\theta = (b, w_1, \ldots, w_n)^T$: every parameter in one column, the intercept first
- X: one row per house; a column of 1s, then one column per feature
- So $X\theta$ gives $b + w_1x_1 + \ldots + w_nx_n$ for every house at once: all m predictions in one product
- The cost becomes $J(\theta)$, and the next slide solves for θ

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The normal equation

$$\theta = (X^\top X)^{-1} X^\top y$$

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The same equation, by hand

Quantity	Value
the three houses	80, 120, 200 m² → 240k, 320k, 540k
X, and θ	a column of 1s (for b) beside the areas; θ = (b, w)
X^TX	rows (3, 400) and (400, 60,800)
X^Ty	(1,100, 165,600)
determinant	22,400
slope w	**2,536 €/m²** (truth 2,400)
intercept b	28,600 €

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And it really is the minimum

- The second derivative is X^TX, never negative in any direction
- So the cost is **convex**: one bottom, no local traps
- Any stationary point (zero gradient) is *the* answer, not *an* answer
- In code: `LinearRegression().fit(X, y)` solves the same problem

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So why learn anything else?

- $(X^TX)^{-1}$ costs about n^3 operations
- Two columns carrying one fact → **no inverse exists**
- Nearly-redundant columns → an inverse that is useless

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How stretched is the valley?

The **condition number** κ = steepest curvature ÷ shallowest curvature: how many times steeper the steepest direction of the bowl is than the shallowest. The curvatures are the eigenvalues of X^TX, so $\kappa = (\texttt{np.linalg.cond(X)})^2$.

- Housing features as recorded: **428,000**
- The same features, standardised: **11.4**
- Add `area_sqft` beside `area_sqm`: **5.2 million**

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The picture behind gradient descent

You are on a hillside in fog.

You cannot see the valley, but you can feel which way the ground slopes.

So: step against the slope, by an amount set by the **learning rate** α, and repeat until the cost stops falling.

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The gradients: one for w, one for b

$$\frac{\partial J}{\partial w} = \frac{1}{m}\sum_i \left(\hat{y}^{(i)} - y^{(i)}\right)x^{(i)}, \qquad \frac{\partial J}{\partial b} = \frac{1}{m}\sum_i \left(\hat{y}^{(i)} - y^{(i)}\right)$$

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One iteration, then the next

1. Predict with the current w and b: $\hat{y}^{(i)} = w \cdot x^{(i)} + b$
2. Measure every error: $e^{(i)} = \hat{y}^{(i)} - y^{(i)}$
3. Average them into the two gradients of the previous slide
4. Step both parameters: $w \leftarrow w - \alpha \cdot \partial J/\partial w$ and $b \leftarrow b - \alpha \cdot \partial J/\partial b$
5. Back to 1: new w and b give new predictions, so new errors

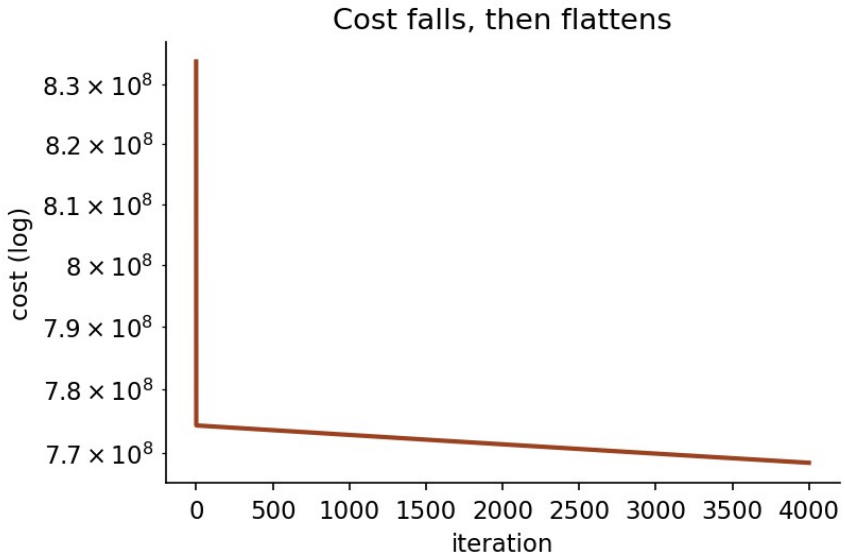
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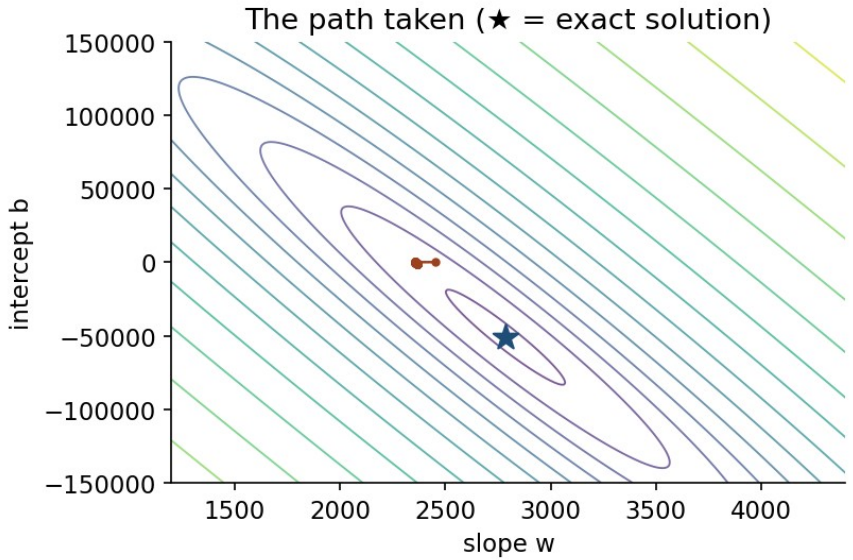
One step on two houses, from w = 0 and b = 0

80 m^2 at 240k, 200 m^2 at 540k; $\alpha = 10^{-5}$	Gradient	After one step
slope w	−63,600	0.636
intercept b	−390	0.0039

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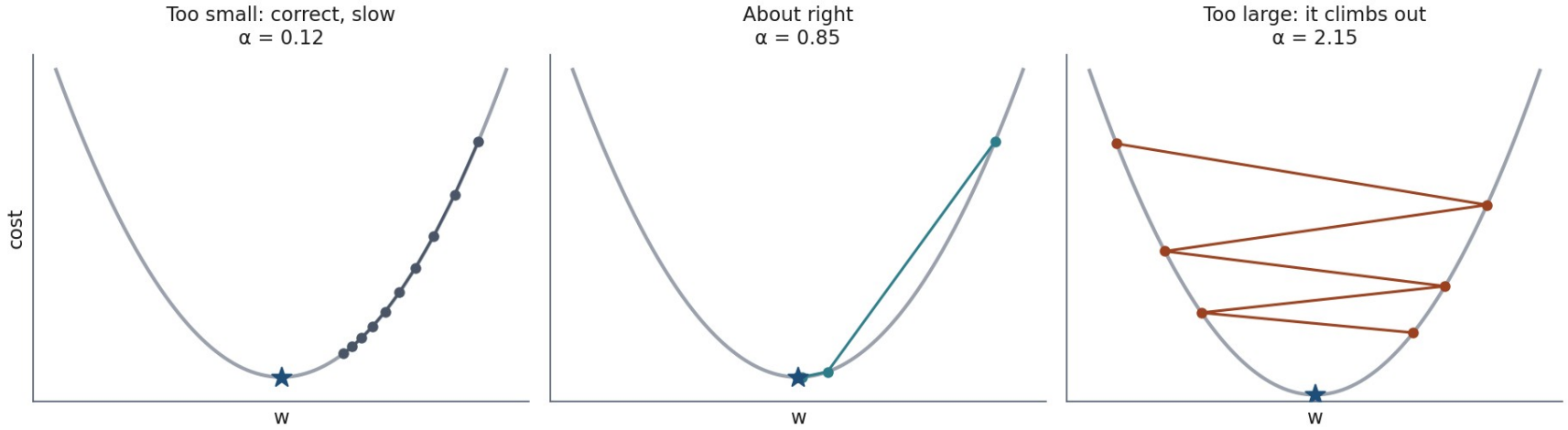
The path it takes





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Three learning rates



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Notebook 1, live

section, and its slides	you call (→ the number to come back with)
one feature, 4-5	`train_test_split` → 450 houses to fit, 150 to test
the exact answer, 15-19	the normal equation in NumPy, then `LinearRegression` → 2,785 €/m², agreeing to 1.82e-12
the iterative answer, 22-27	NumPy only: gradient descent → the same answer, slowly
all six features, 52-53	`LinearRegression` → area 2,421, bedrooms −16.6%
the baseline, 12	`DummyRegressor(strategy="mean")`, `r2_score` → 96,440 € against 20,341 €, R^2 0.956

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Break

What if the relationship bends?

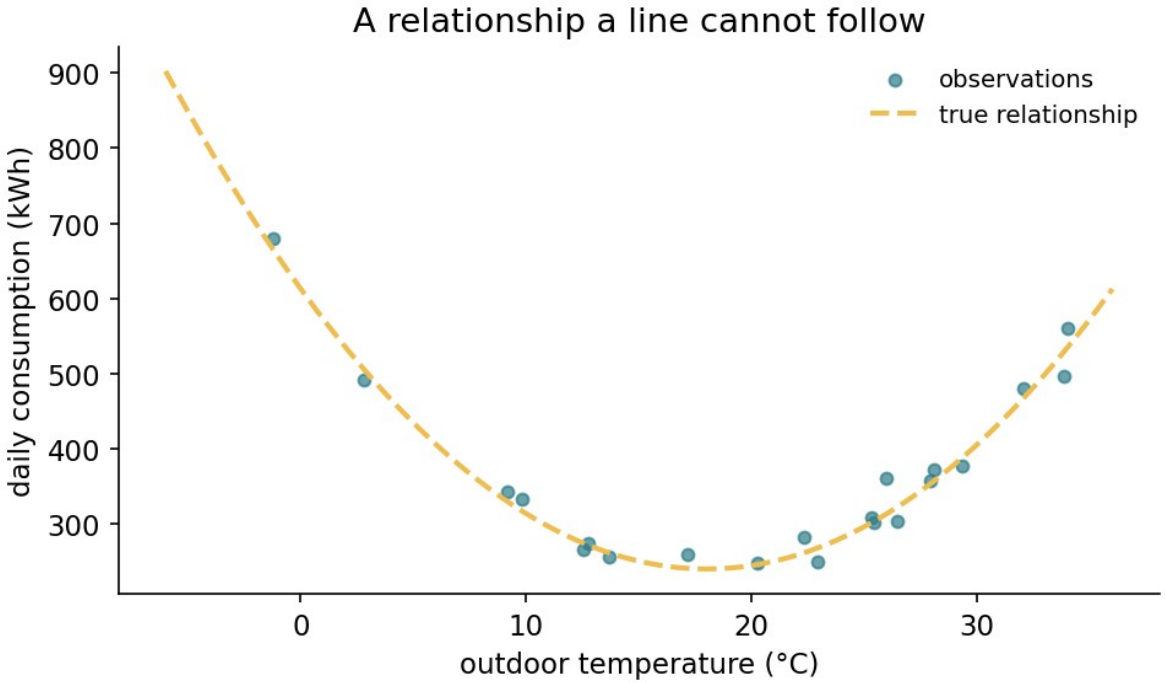
Energy consumption against temperature: heating in the cold, cooling in the heat.

No straight line follows that.

In code: `PolynomialFeatures(degree=2)` adds the squared temperature, t^2, as a second column.

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A relationship a line cannot follow



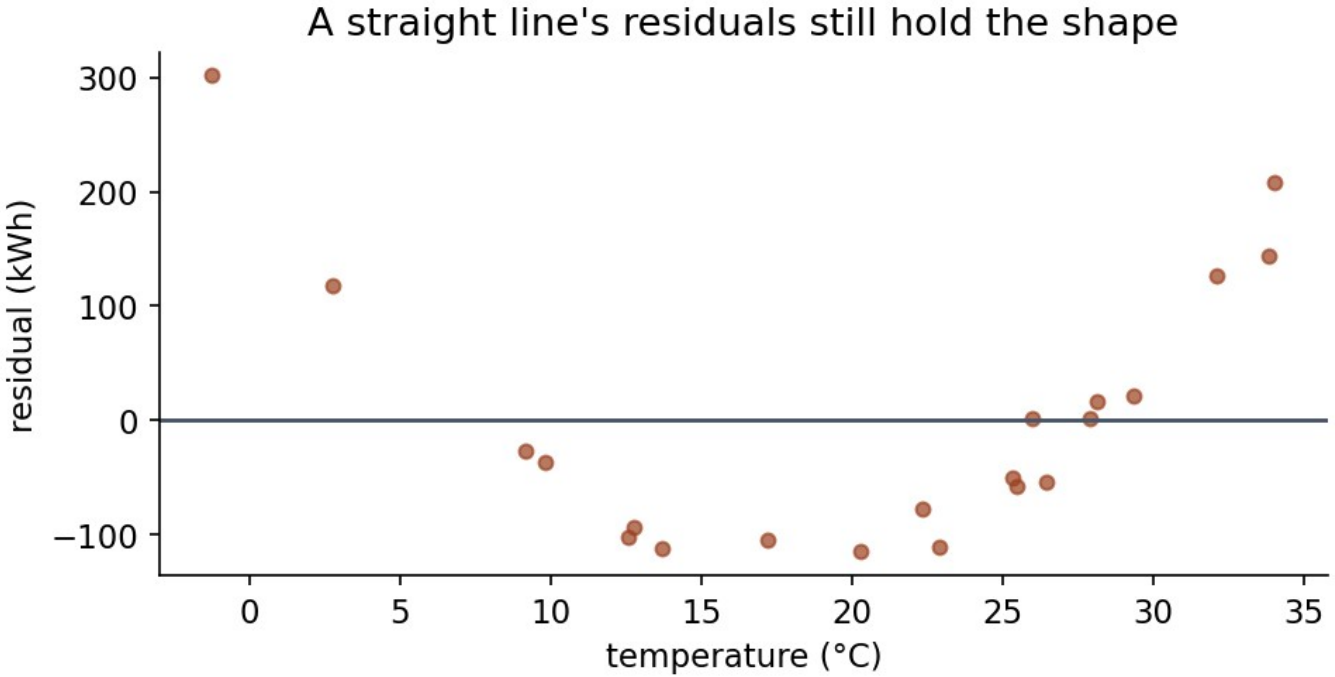
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“Linear” means linear in the coefficients

$$\hat{y} = w_1 t + w_2 t^2 + b$$

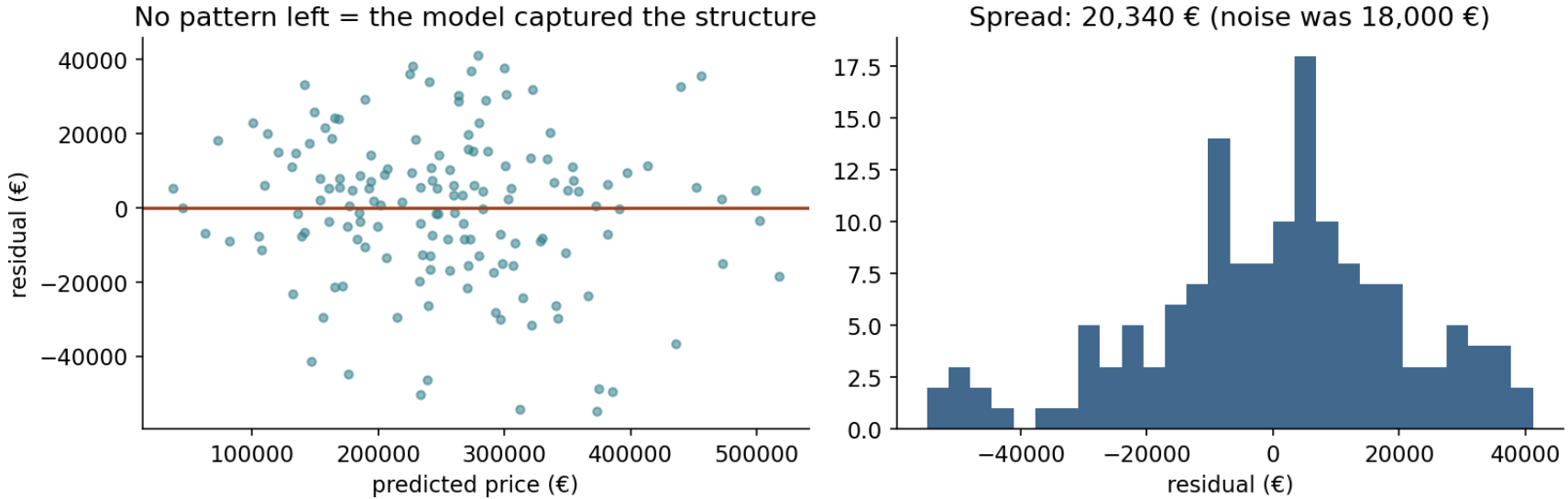
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The straight line’s residuals keep the shape



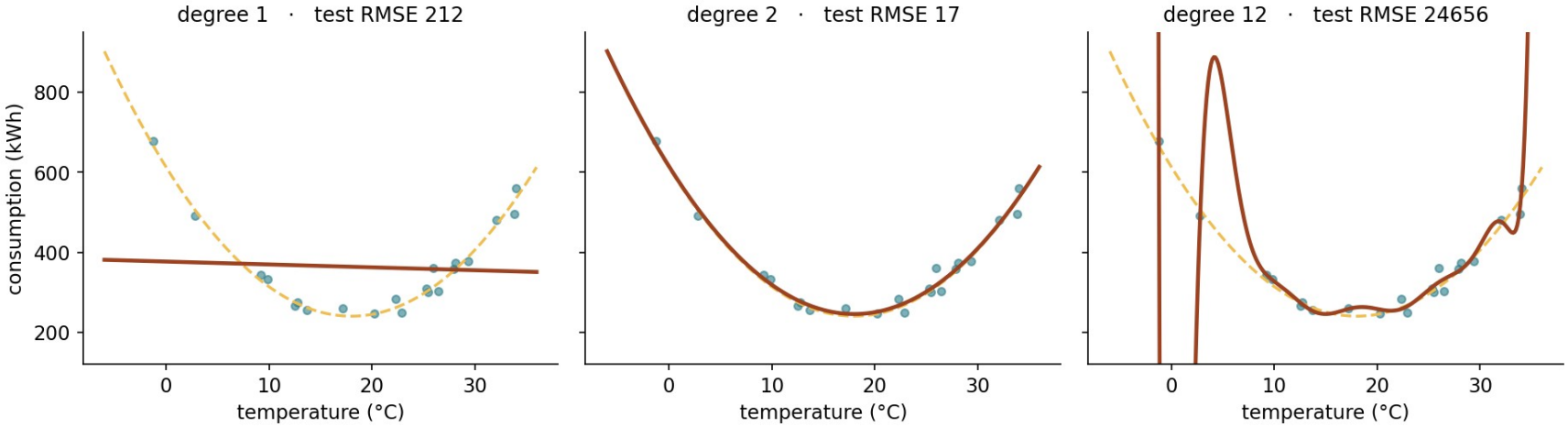
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What “nothing left” looks like



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So use more flexibility?



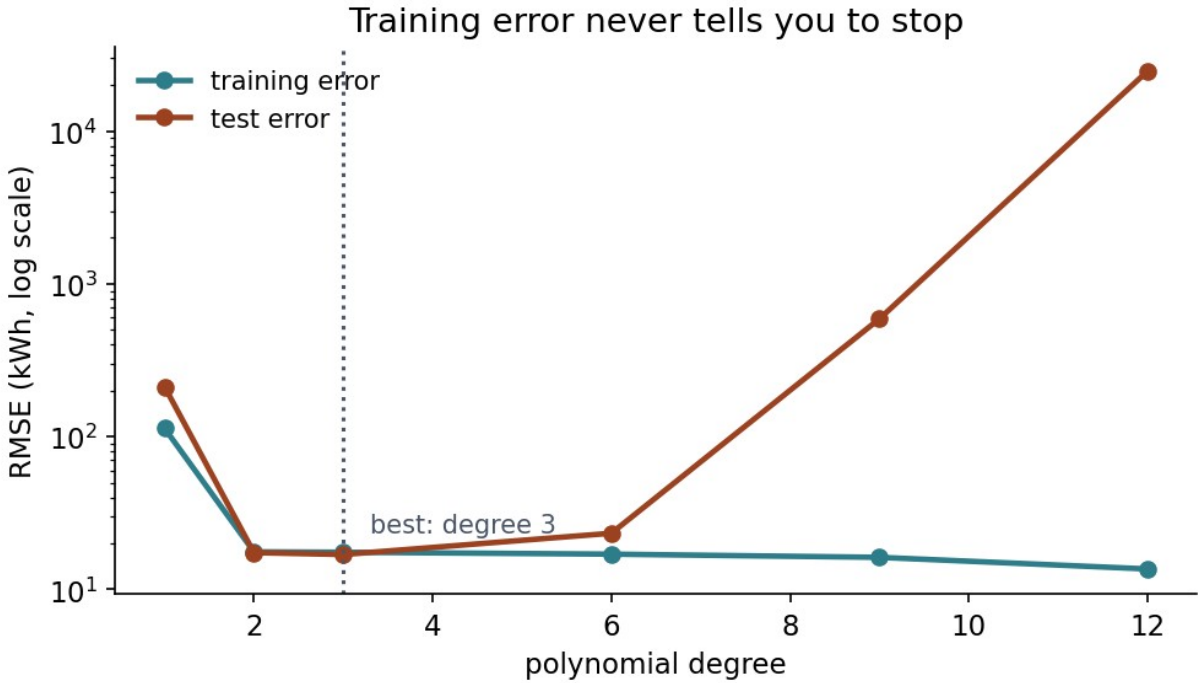
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Root mean squared error (RMSE), by degree

Degree	Train RMSE (kWh)	Test RMSE (kWh)
1	113.3	212.1
3	17.5	**16.9**
9	16.2	590.1
12	13.6	24,655.7

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The gap is the overfitting



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Notebook 2, live

section, and its slides	you call (→ the number to come back with)
a relationship that bends, 30-31	`train_test_split` → 21 days to fit, 9 to test
linear in the coefficients, 32-34	`PolynomialFeatures`, `StandardScaler` and `LinearRegression`, chained by `make_pipeline`
too much flexibility, 35-37	`mean_squared_error`, square-rooted → test RMSE 16.9 at degree 3, 24,655.7 at 12
why the wiggles appear, 39	the fitted coefficients → largest 3,097,038,010 at degree 12

Why flexibility goes wrong

Largest coefficient, degree 2: **411**

Largest coefficient, degree 12: **3,097,038,010**

Charge for size

$$J_{\text{ridge}} = \text{MSE} + \lambda \sum_j w_j^2 \qquad J_{\text{lasso}} = \text{MSE} + \lambda \sum_j |w_j|$$

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Two rules that are not optional

- The **intercept is never penalised**: it is not a claim about any feature
- Features **must be scaled first**, or the penalty is arbitrary
- In code: `make_pipeline(StandardScaler(), Ridge(alpha=0.01))`; scikit-learn calls λ `alpha`

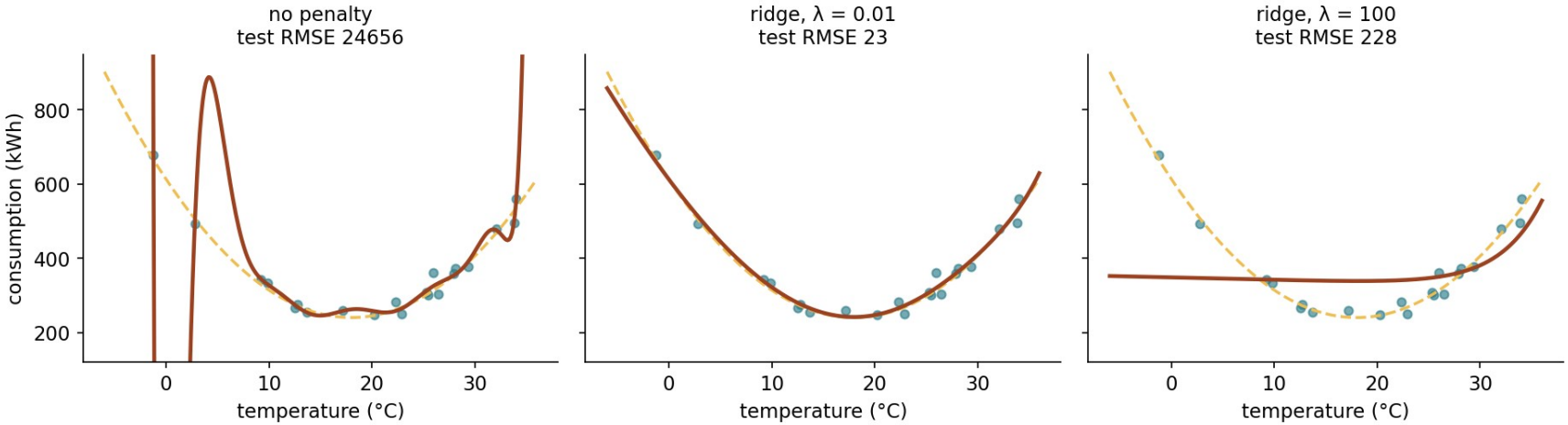
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What a penalty buys

Model	Train RMSE	Test RMSE	max |w|
none	13.6	24,655.7	3,097,038,010
$\lambda = 0.01$	18.0	**22.8**	365
$\lambda = 1$	44.5	105.2	156
$\lambda = 100$	96.0	227.6	6

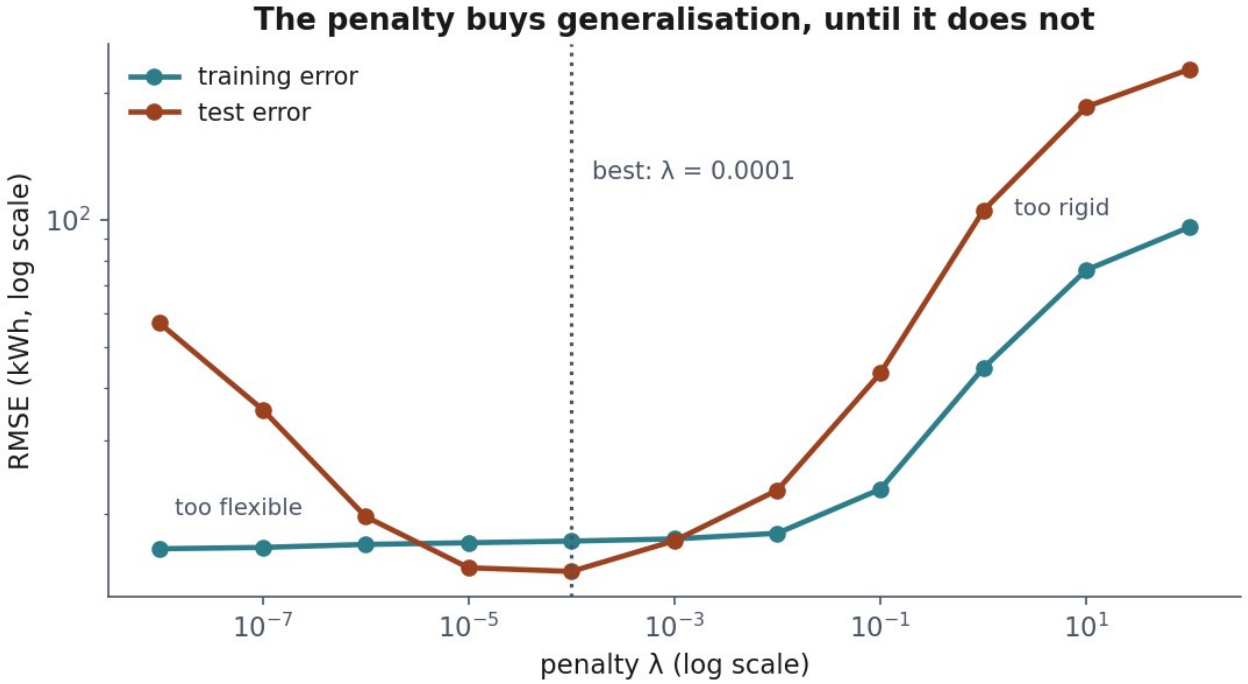
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The unpenalised fit leaves the picture



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Too flexible, too rigid



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Ridge has an exact solution too

$$\theta = (X^\top X + \lambda I)^{-1} X^\top y$$

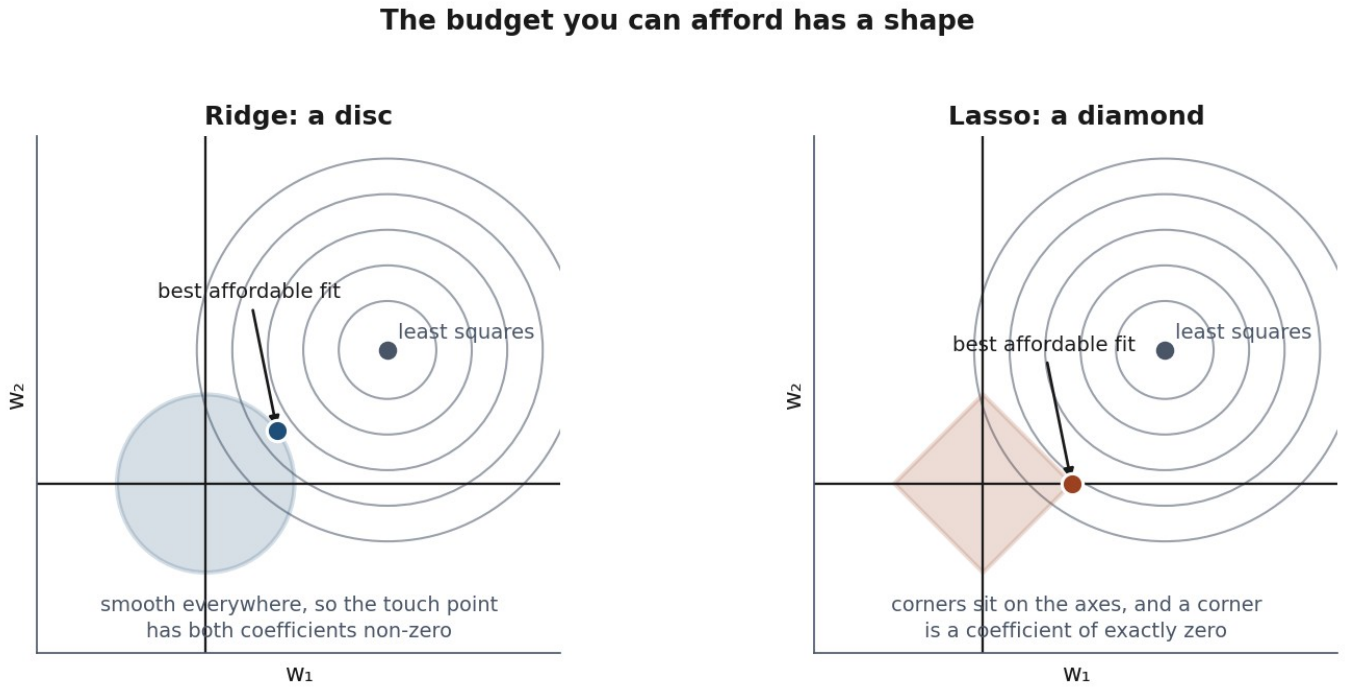
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Why that one change fixes it

- Least squares fails when a direction costs nothing: the valley is flat
- The penalty makes **every** direction cost something
- Every eigenvalue of X^TX shifts up by λ, so none of them can be zero

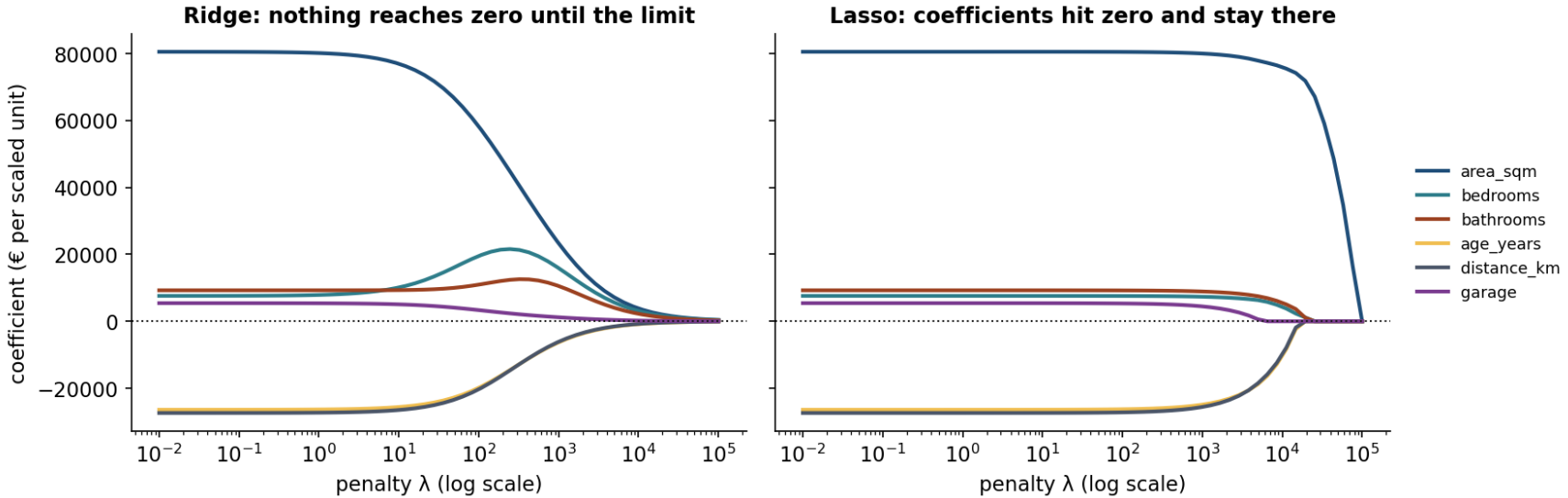
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The picture behind Ridge vs Lasso



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Ridge shrinks, Lasso selects



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The order Lasso drops features

λ	Features kept
1,000	all six
10,000	five: `garage` dropped
20,000	three: `area_sqm`, `bedrooms`, `bathrooms`
40,000	one: `area_sqm`

The case Ridge was invented for

`area_sqm` and `area_sqft`: the same measurement, two units.

Correlation: **0.999999**

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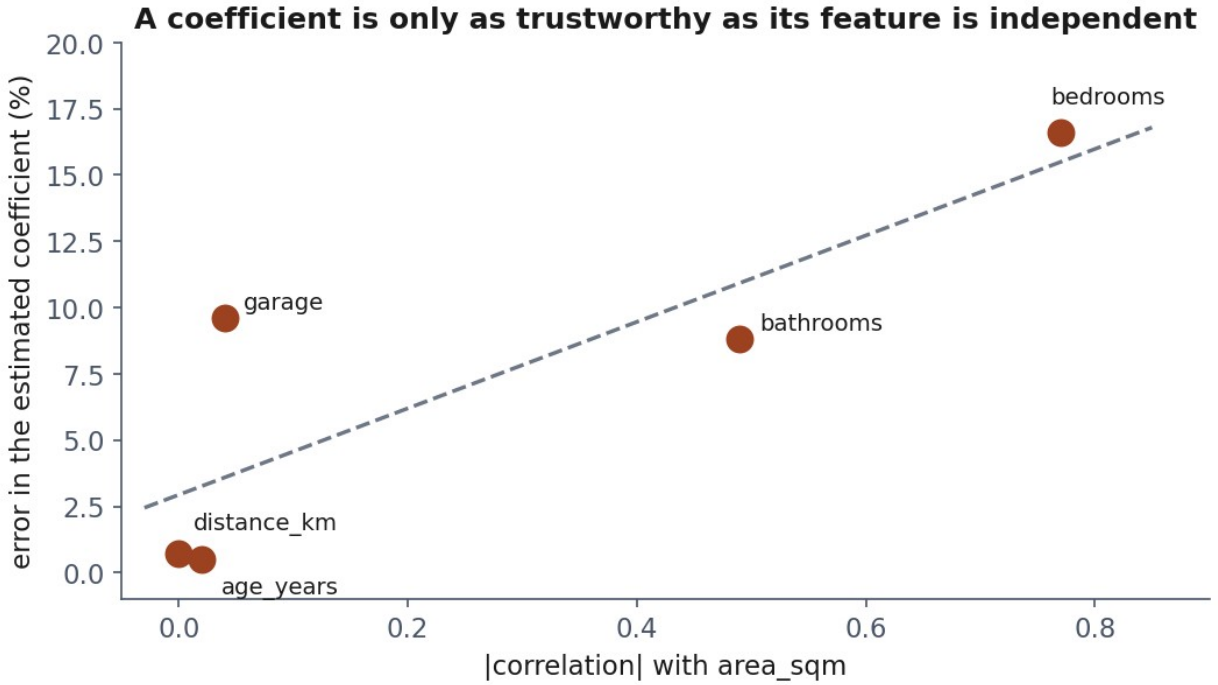
The coefficients are noise

Split	area_sqm	area_sqft
0	9,261	−640
1	14,535	−1,127
2	45,307	−3,989
3	39,061	−3,402

What a coefficient actually says

The change in the prediction for a one-unit change in that feature, **with every other feature held fixed**.

When can you trust a coefficient?



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Notebook 3, live

section, and its slides	you call (→ the number to come back with)
Ridge on the degree-12 fit, 40-44	`Ridge(alpha=0.01)` after `PolynomialFeatures` and `StandardScaler` → largest coefficient 365
regularisation paths, 47-49	`make_pipeline(StandardScaler(), Ridge(alpha=...))`, the same with `Lasso` → only area left at 40,000
the collinear case, 50-51	`LinearRegression` against `Ridge(alpha=10)` on four splits → 9,261 to 45,307, then about 40,000

What we did today

- A model that claims a fixed amount per unit
- An exact solution (a projection), and when it fails
- An iterative one: walk downhill, mind your stride
- Flexibility helps until it does not
- Charging for size buys generalisation, and readable coefficients
- **365** against billions: what a penalty of 0.01 costs, and buys

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Homework: we discuss it on Friday 23 October

`Exercises/03_regression.md`

Fit, regularise, and **explain which coefficients you believe**.

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